

Qiong Wang

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HIGHLIGHTS

- ▷ **Passion for AI4Healthcare & AI4Neuroscience** – 3 papers on biomedical image analysis (ACR, IEEE JBHI)
- ▷ **Passion for Real-World Tech Applications** – Top tech company (Tiktok) & 2 Startups (acquired for RMB 700K)
- ▷ **Passion for Art** – 'Blue-and-White Porcelain Vessel Design' Outstanding Graduation Project

EDUCATION

Boston University Sep 2023 – May 2025
Master of Science in Computer Science - GPA: 3.51/4.00 Boston, US

Zhengzhou University Sep 2007 – June 2011
Bachelor of Arts in Art Design - Ranked 1st in Graduation Project Zhengzhou, China

PUBLICATIONS

Enhancing Hand Osteoarthritis Classification with Generative AI: A CycleGAN and EfficientNetB7 Approach

- Zhen Cao, Juan Shan, Xiaohan Jiang, **Qiong Wang**, Timothy McAlindon, Jeffrey Driban, Ming Zhang
- *The American College of Rheumatology (ACR) Annual Meeting 2025. Accepted.*

Enhancing Bone Marrow Lesion Segmentation through Dual-Channel Deep Neural Networks and Test-Time Augmentation

- Shihua Qin, **Qiong Wang**, Juan Shan, Jeffery Driban, Timothy McAlindon, Kevin Wang, Ming Zhang
- *IEEE Journal of Biomedical and Health Informatics. Under Review.*

Optimized Deep Learning Method for Automated Segmentation of Bone Marrow Lesions

- Shihua Qin, **Qiong Wang**, Juan Shan, Jeffrey Driban, Timothy McAlindon, Kevin Wang, Ming Zhang
- *The Osteoarthritis Research Society International (OARSI) 2025 Conference. Accepted.*

A Novel Machine Learning Model to Predict Knee Replacement (*Manuscript in preparation*)

- **Qiong Wang**, Juan Shan, Ming Zhang

RESEARCH PROJECT

Multi-Model Pipeline for 3D Neuronal Mitochondria Segmentation and Proofreading in EM Connectomics Jun 2025 – Present
Boston College

DVISORS: Prof. Donglai Wei

- Implemented semantic segmentation using nnUNet, and utilized an instance segmentation framework combining SAM, watershed, and cc3d to delineate neuronal mitochondria at scale. Applied Cellable 3D for proofreading and refinement.

Mitochondria Classification in H01 Connectomics Dataset using 3D ResNets May 2025 – Jul 2025
Boston College

DVISORS: Prof. Donglai Wei; Collaborator: Prof. Eva Anton (UNC School of Medicine)

- Implemented 3D ResNet-based pipeline for proofreading H01 E-I neuron pair mitochondria, benchmarking multiple architectures (ResNet18/50, 2.5D/3D/ACS) with MedMNIST3D, and achieving robust performance (Acc 0.91).

Enhancing Hand Osteoarthritis Classification with Generative AI: A CycleGAN and EfficientNetB7 Approach Sep 2024 – May 2025
AICV Lab

ADVISOR: Prof. Ming Zhang, Prof. Juan Shan

- Developed a CycleGAN pipeline to generate severe OA images (KL3/4) from mild X-rays (KL0/1) and integrated them into EfficientNetB7, achieving 6.0% and 3.1% accuracy improvements for KL3 and KL4 classification.

Enhancing BML Segmentation through Dual-Channel Deep Neural Networks and Test-Time Augmentation Jan 2024 – Mar 2025
AICV Lab

ADVISOR: Prof. Ming Zhang

- Implemented the dual channel and TTA pipeline for BML segmentation on MRI images, evaluating deep learning models (Residual U-Net, SwinUNetR, AttentionUNet, U-Net++) and achieving a 69.0% Dice score with U-Net++.

Transformer-based Retrieval and Generation for QA with DistilBERT Encoder and Mistral-7B Decoder Apr 2025
Boston University

ADVISOR: Prof. Mikhail Chertushkin

- Developed a Transformer-based QA system with DistilBERT (encoder-only) for retrieval and Mistral-7B (decoder-only) with LoRA fine-tuning for generation, evaluated on IR and BLEU metrics with score: 0.38.

Neural Machine Translation with Luong Attention for Sequence Alignment <i>ADVISOR: Prof. Mikhail Chertushkin</i>		2025 <i>Boston University</i>
Implemented a Seq2Seq model with Luong attention for sequence alignment, using attention-weighted decoding, teacher forcing, and sequence-level loss optimization, achieving a BLEU score of 30.26.		
Breast Cancer Detection via Attention-Enhanced ImprovedUNet for Multi-Class Breast Ultrasound Image Segmentation <i>ADVISOR: Prof. Mikhail Chertushkin</i>		Feb 2025 <i>Boston University</i>
Developed an attention-enhanced UNet for multi-class breast ultrasound segmentation, integrating attention within skip connections, optimized with AdamW and cosine annealing LR scheduling, improving weighted Dice from 0.54 to 0.79.		
RESEARCH EXPERIENCE		
Research Assistant, Boston College <i>Advisor: Prof. Donglai Wei</i>		May 2025 – Present
Research Assistant, AICV Lab at Boston University <i>Advisor: Prof. Ming Zhang, Prof. Juan Shan</i>		Jan 2024 – May 2025
INDUSTRY EXPERIENCE		
ByteDance Technology Co. LTD (TikTok) Product Manager – Led Machine Learning Model Video Search Project, and AI Video Batch Editing Project.		Feb 2021 – Dec 2022 <i>Shanghai, China</i>
The Little Black Card APP Product Manager & Backend Engineer – Founding team member; led product development, reaching RMB 92M monthly GMV and securing Series B financing. Built a referral-based distribution system with over 90% user retention, ranked top 3 in China in 2019.		Jan 2016 – Jan 2021 <i>Shanghai, China</i>
Vivian Pearl (E-commerce Startup) Founder & Software Engineer – Founded a jewelry brand, scaled sales via online channels, and acquired in 2015 for RMB 700K.		Jan 2014 – Oct 2015 <i>Beijing, China</i>
Infinite Travel (Mobile App Startup) Founder & Software Engineer – Built a hotel reservation app, led a 7-member team, and secured RMB 200K angel funding.		May 2013 – Oct 2015 <i>Beijing, China</i>
Sohu.com Limited (Top Chinese Tech Company (2011)) User Interface (UI) Designer		Sep 2011 – Apr 2013 <i>Beijing, China</i>
HONORS & AWARDS		
The Seed Research Grant		<i>Boston University, 2024 – 2025</i>
Outstanding Project Achievement Award		<i>TikTok, Oct 2022</i>
Permanent Member, ByteDance Strategic Advisory Committee		<i>TikTok, 2021 – 2022</i>
Excellence in Individual Contribution Award		<i>TikTok, Nov 2021</i>
Achieved Acquisition of Vivian Pearl Brand for 700,000 RMB		<i>Oct 2015</i>
Raised 200,000 RMB in Angel Investment for Infinite Travel APP		<i>Dec 2013</i>
GRADUATE-LEVEL COURSES		
Theory CS566 Analysis of Algorithm (<i>Teaching Assistant, Fall 2024 & Spring 2025</i>) CS662 Computer Language Theory		
System CS472 Computer Architecture CS575 Operating Systems CS579 Database Management CS665 Software Design and Patterns		
Machine Learning CS555 Foundation of Machine Learning CS677 Data Science with Python CS767 Advanced Machine Learning and Neural Networks		